DECENT LOCALLY SEPARATED SPACES HAVE CLOSED DIAGONALS IN THE FIBRES OVER POINTS

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span; the diagonal-fibre identification is delegated to an uncarded chapter lemma and declared external.

Allowed standard prerequisites (editorial).

- Fibres of the diagonal. lemma-fibres-diagonal (same chapter, NOT carded, declared external). Identifies the fibre of the diagonal with a scheme fibre.
- Points of the presentation. Properties of Spaces, lemma points-presentation (cross-chapter, named). The two points u, u' of U over x.

Source reference. Original lemma 03JU: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 534–534. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 088I: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 3675–3675. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04Q9: focused-seeds/geometry/S9-decent-locally-separated-decent/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original remark 03II: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 459–459. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01IQ: focused-seeds/geometry/S9-decent-locally-separated-decent/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03DP: focused-seeds/geometry/S9-decent-locally-separated-decent/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted

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proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03KN: focused-seeds/geometry/S9-decent-locally-separated-decent/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03BS: focused-seeds/geometry/S9-decent-locally-separated-decent/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1 (David Rydh)**.** *A locally separated algebraic space is decent.*

Proof. Let S be a scheme and let X be a locally separated algebraic space over S. We may assume $S = \mathrm{Spec}(\mathbf{Z})$, see Properties of Spaces, Definition 03BS. Unadorned fibre products will be over $\mathbf{Z}$. Let $x \in |X|$. Choose a scheme U, an étale morphism $U \to X$, and a point $u \in U$ mapping to x in $|X|$. As usual we identify $u = \mathrm{Spec}(\kappa(u))$. As X is locally separated the morphism

$$u \times_X u \to u \times u$$

is an immersion (Morphisms of Spaces, Lemma 03KN). Hence More on Groupoids, Lemma 04Q9 tells us that it is a closed immersion (use Schemes, Lemma 01IQ). As $u \times_X u \to u \times_X U$ is a monomorphism (base change of $u \to U$) and as $u \times_X U \to u$ is étale we conclude that $u \times_X u$ is a disjoint union of spectra of fields (see Remark 03II and Schemes, Lemma 03DP). Since it is also closed in the affine scheme $u \times u$ we conclude $u \times_X u$ is a finite disjoint union of spectra of fields. Thus x can be represented by a monomorphism $\mathrm{Spec}(k) \to X$ where k is a field, see Lemma 03JU.

Next, let $U = \mathrm{Spec}(A)$ be an affine scheme and let $U \to X$ be an étale morphism. To finish the proof it suffices to show that $F = U \times_X \mathrm{Spec}(k)$ is finite. Write $F = \coprod_{i \in I} \mathrm{Spec}(k_i)$ as the disjoint union of finite separable extensions of k. We have to show that I is finite. Set $R = U \times_X U$. As X is locally separated, the morphism $j : R \to U \times U$ is an immersion. Let $U' \subset U \times U$ be an open such that j factors through a closed immersion $j' : R \to U'$. Let $e : U \to R$ be the diagonal map. Using that e is a morphism between schemes étale over U such that $\Delta = j \circ e$ is a closed immersion, we conclude that $R = e(U) \amalg W$ for some open and closed subscheme $W \subset R$. Since j' is a closed immersion we conclude that $j'(W) \subset U'$ is closed and disjoint from $j'(e(U))$. Therefore $\overline{j(W)} \cap \Delta(U) = \emptyset$ in $U \times U$. Note that W contains $\mathrm{Spec}(k_i \otimes_k k_{i'})$ for all $i \neq i'$, $i, i' \in I$. By Lemma 088I we conclude that I is finite as desired. $\square$